New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

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Business Questions

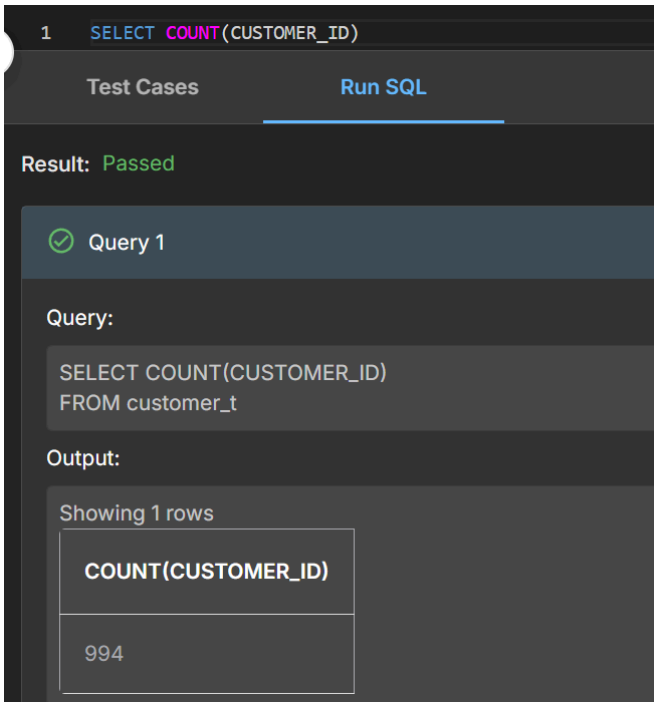
Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
/* Total number of customers */
SELECT
    COUNT(customer_id) AS Total_Customers
FROM customer_t;

/* Number of customers by state */
SELECT
    state,
    COUNT(customer_id) AS Number_of_Customers
FROM customer_t
GROUP BY state
ORDER BY Number_of_Customers DESC;
```

Output:



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Query 1

Query:

```
SELECT
   state,
   COUNT(customer_id) AS Number_of_Customers
FROM customer_t
GROUP BY state
ORDER BY Number_of_Customers DESC
```

Output:

Showing first 10 rows out of 49 rows

state	Number_of_Customers
Texas	97
California	97
Florida	86
New York	69
District of Columbia	35
Ohio	33
Colorado	33
Alabama	29

Observations and Insights:

- The total number of customers is 994.

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- The top five states with maximum number of customers are: Texas, California, Florida, New York and District of Columbia.
- The bottom 3 states with the minimum number of customers are: Maine, Vermont, Wyoming with 1 customer only.

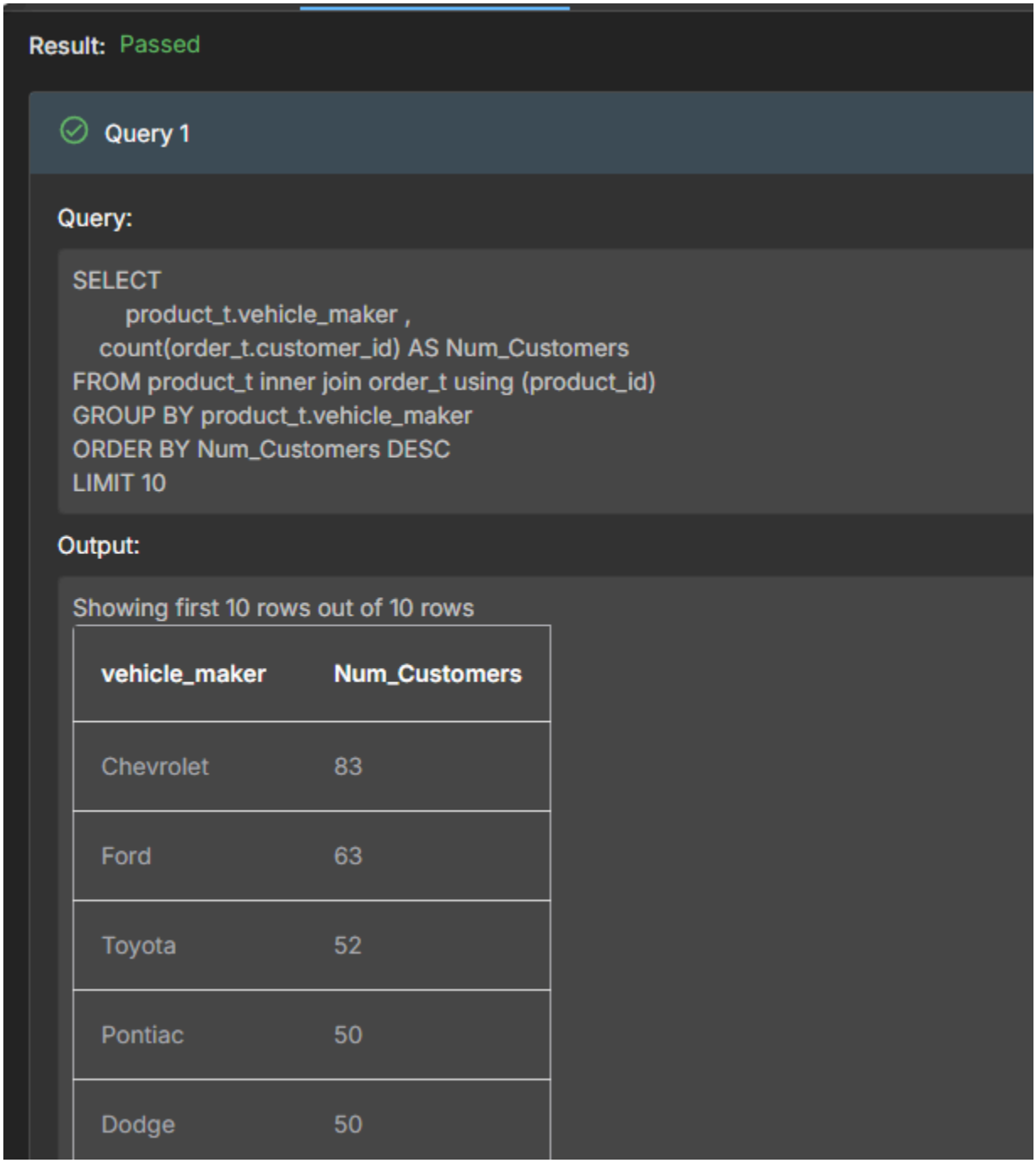
Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT
        product_t.vehicle_maker ,
   count(order_t.customer_id) AS Num_Customers
FROM product_t inner join order_t using (product_id)
GROUP BY product_t.vehicle_maker
ORDER BY Num_Customers DESC
LIMIT 10
```

Output:

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- 
Result: Passed

Query 1

Query:

```
SELECT
      product_t.vehicle_maker ,
   count(order_t.customer_id) AS Num_Customers
FROM product_t inner join order_t using (product_id)
GROUP BY product_t.vehicle_maker
ORDER BY Num_Customers DESC
LIMIT 10
```

Output:

Showing first 10 rows out of 10 rows

vehicle_maker	Num_Customers
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations and Insights:

- Top 5 vehicle makers with highest number of customers are: Chevrolet, Ford, Toyota, Pontiac and Dodge
- Bottom 5 vehicle makers with only 1 customer are: Austin, Citron, Daewoo, MG, Ram
- Need to check the revenue generated by the vehicle makers with only 1 customer.

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Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT
   state,
   vehicle_maker,
   customer_count
FROM (
   SELECT
      state,
      vehicle_maker,
      COUNT(customer_id) AS customer_count,
      RANK() OVER (
         PARTITION BY state
         ORDER BY COUNT(customer_id) DESC
      ) AS Rnk
   FROM order_t O
   INNER JOIN product_t P USING (product_id)
   INNER JOIN customer_t C USING (customer_id)
   GROUP BY state, vehicle_maker
) AS Temp
WHERE Rnk = 1
ORDER BY state;
```

Output:

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Result: Passed

Query 1

Query:

```
SELECT
   state,
   vehicle_maker,
   customer_count
FROM (
   SELECT
      state,
      vehicle_maker,
      COUNT(customer_id) AS customer_count,
      RANK() OVER (
         PARTITION BY state
         ORDER BY COUNT(customer_id) DESC
      ) AS rnk
   FROM order_t O
   INNER JOIN product_t P USING (product_id)
   INNER JOIN customer_t C USING (customer_id)
   GROUP BY state, vehicle_maker
) AS Temp
WHERE rnk = 1
ORDER BY state
```

Output:

Showing first 10 rows out of 143 rows

state	vehicle_maker	customer_count
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arizona	Cadillac	3

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Observations and Insights:

- In some states, there are multiple most preferred vehicle makers with rank=1 as they have the same number of customers.
- Sorting in descending order by count of customers shows that-
    - Texas has Chevrolet with max customer count- 9
    - Florida has Toyota with max customer count=7
    - California has Nissan, Ford, Dodge, Chevrolet, Audi all with customer count=6
- All states have variation in who is the #1 preferred vehicle maker, but Chevrolet is a common preferred customer.

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Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

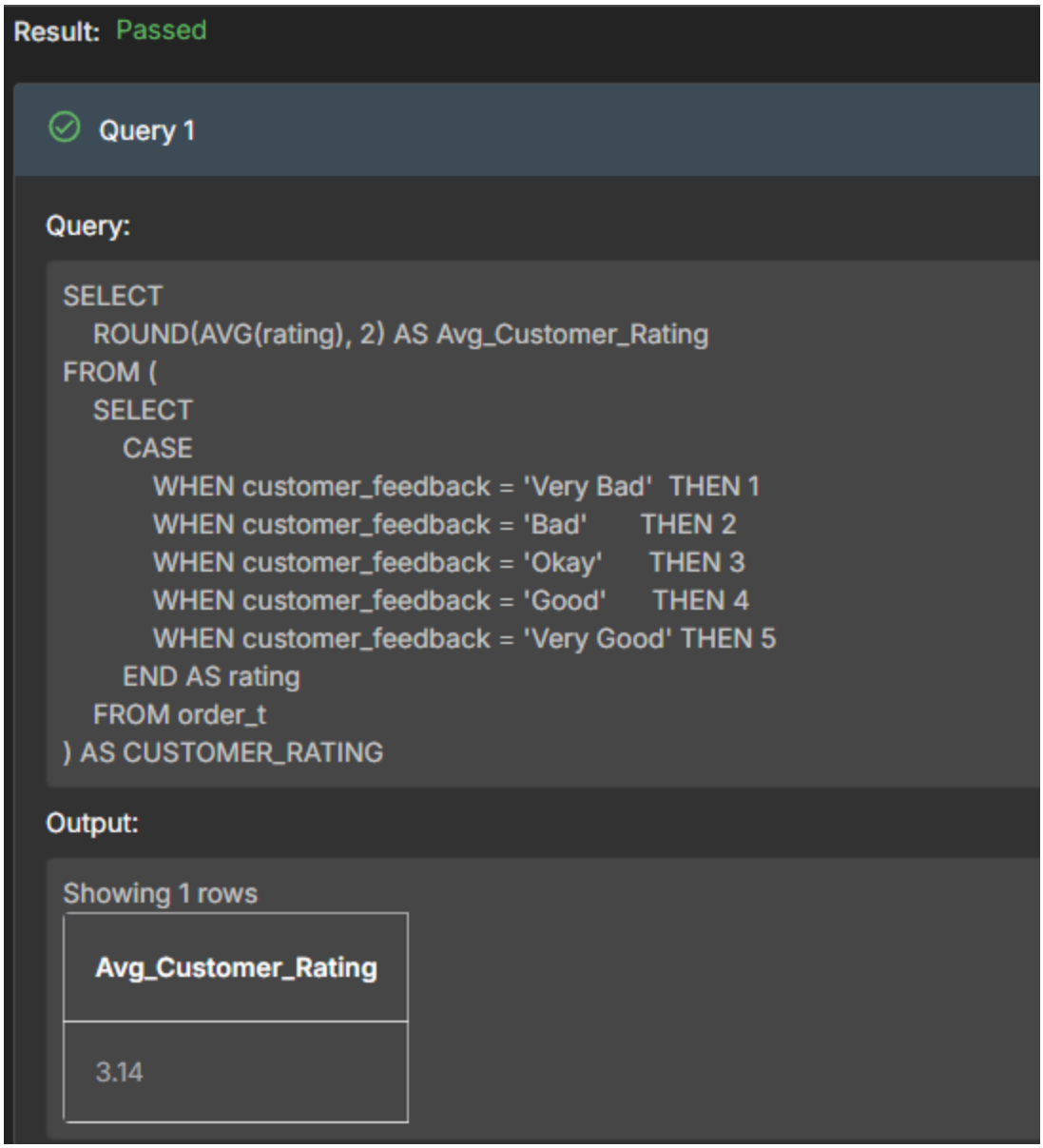
Solution Query:

Query 1: Overall average rating given by customers

```
SELECT
   ROUND(AVG(rating), 2) AS Avg_Customer_Rating
FROM (
   SELECT
      CASE
         WHEN customer_feedback = 'Very Bad'  THEN 1
         WHEN customer_feedback = 'Bad'       THEN 2
         WHEN customer_feedback = 'Okay'      THEN 3
         WHEN customer_feedback = 'Good'      THEN 4
         WHEN customer_feedback = 'Very Good' THEN 5
      END AS rating
   FROM order_t
) AS CUSTOMER_RATING
```

Output:

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Query 2:

```
SELECT
   'Q' || quarter_number AS Quarter,
   ROUND(AVG(rating), 2) AS Avg_Customer_Rating
FROM (
   SELECT
      quarter_number,
      CASE
         WHEN customer_feedback = 'Very Bad'  THEN 1
```

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```
        WHEN customer_feedback = 'Bad'      THEN 2
        WHEN customer_feedback = 'Okay'     THEN 3
        WHEN customer_feedback = 'Good'     THEN 4
        WHEN customer_feedback = 'Very Good' THEN 5
    END AS rating
  FROM order_t
) AS CUSTOMER_RATING
GROUP BY quarter_number
```

Output:

Result: Passed

Query 1

Query:

```
SELECT
   'Q' || quarter_number AS Quarter,
   ROUND(AVG(rating), 2) AS Avg_Customer_Rating
FROM (
   SELECT
      quarter_number,
      CASE
         WHEN customer_feedback = 'Very Bad'  THEN 1
         WHEN customer_feedback = 'Bad'       THEN 2
         WHEN customer_feedback = 'Okay'      THEN 3
         WHEN customer_feedback = 'Good'      THEN 4
         WHEN customer_feedback = 'Very Good' THEN 5
      END AS rating
   FROM order_t
) AS CUSTOMER_RATING
GROUP BY quarter_number
ORDER BY Avg_Customer_Rating DESC
```

Output:

Showing 4 rows

Quarter	Avg_Customer_Rating
Q1	3.55
Q2	3.35
Q3	2.96
Q4	2.4

Observations and Insights:

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- The overall average rating is 3.14
- The average rating has decreased over the year from Q1-3.55 (very good) to Q4 2.4 (OK)
- Customer satisfaction has gone down through the year.

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Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT
   'Q' || quarter_number AS Quarter,
   ROUND((Very_Bad   * 100.0) / Total_Feedback, 2) AS Perc_Very_Bad,
   ROUND((Bad        * 100.0) / Total_Feedback, 2) AS Perc_Bad,
   ROUND((Okay       * 100.0) / Total_Feedback, 2) AS Perc_Okay,
   ROUND((Good       * 100.0) / Total_Feedback, 2) AS Perc_Good,
   ROUND((Very_Good  * 100.0) / Total_Feedback, 2) AS Perc_Very_Good
FROM (
   SELECT
      quarter_number,
      SUM(CASE WHEN customer_feedback = 'Very Bad'  THEN 1 ELSE 0 END) AS Very_Bad,
      SUM(CASE WHEN customer_feedback = 'Bad'       THEN 1 ELSE 0 END) AS Bad,
      SUM(CASE WHEN customer_feedback = 'Okay'      THEN 1 ELSE 0 END) AS Okay,
      SUM(CASE WHEN customer_feedback = 'Good'      THEN 1 ELSE 0 END) AS Good,
      SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) AS Very_Good,
      COUNT(customer_feedback) AS Total_Feedback
   FROM order_t
   GROUP BY quarter_number
) AS FEEDBACK
ORDER BY Quarter;
```

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Output:

Query 1

Query:

```
SELECT
   'Q' || quarter_number AS Quarter,
   ROUND((Very_Bad   * 100.0) / Total_Feedback, 2) AS Perc_Very_Bad,
   ROUND((Bad        * 100.0) / Total_Feedback, 2) AS Perc_Bad,
   ROUND((Okay       * 100.0) / Total_Feedback, 2) AS Perc_Okay,
   ROUND((Good       * 100.0) / Total_Feedback, 2) AS Perc_Good,
   ROUND((Very_Good  * 100.0) / Total_Feedback, 2) AS Perc_Very_Good
FROM (
   SELECT
      quarter_number,
      SUM(CASE WHEN customer_feedback = 'Very Bad'  THEN 1 ELSE 0 END) AS Very_Bad,
      SUM(CASE WHEN customer_feedback = 'Bad'       THEN 1 ELSE 0 END) AS Bad,
      SUM(CASE WHEN customer_feedback = 'Okay'      THEN 1 ELSE 0 END) AS Okay,
      SUM(CASE WHEN customer_feedback = 'Good'      THEN 1 ELSE 0 END) AS Good,
      SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) AS Very_Good,
      COUNT(customer_feedback) AS Total_Feedback
   FROM order_t
   GROUP BY quarter_number
) AS FEEDBACK
ORDER BY Quarter
```

Output:

Showing 4 rows

Quarter	Perc_Very_Bad	Perc_Bad	Perc_Okay	Perc_Good	Perc_Very_Good
Q1	10.97	11.29	19.03	28.71	30
Q2	14.89	14.12	20.23	22.14	28.63
Q3	17.9	22.71	21.83	20.96	16.59
Q4	30.65	29.15	20.1	10.05	10.05

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Observations and Insights:

- Percentage of Good and Very good feedback has decreased over the four quarters. Quarter 4 has the least percentage of good ratings.
- Percentage of Bad and Very bad feedback has increased over the four quarters. Quarter 4 has the highest percentage of dissatisfied customers.
- Customers are definitely getting more dissatisfied over time!

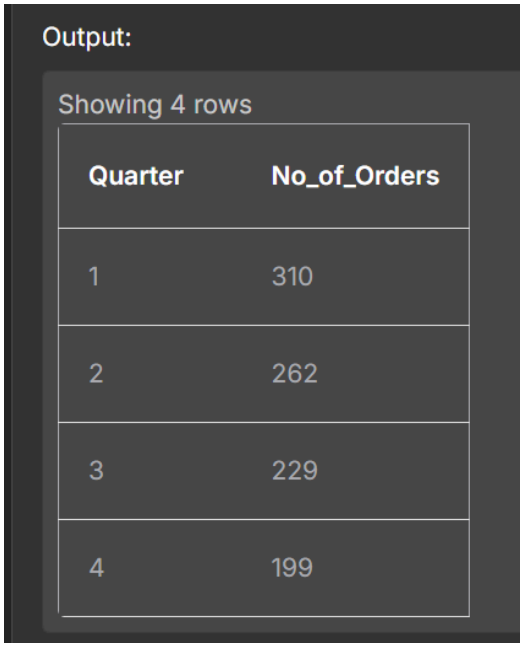
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Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT
   quarter_number AS Quarter,
   COUNT(order_id) AS No_of_Orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:


Output:

Showing 4 rows

Quarter	No_of_Orders
1	310
2	262
3	229
4	199

Observations and Insights:

- The number of orders have decreased over time from 310 in Q1 to 199 in Q4.
- In every quarter, there are less orders than before.
- The number of orders going down correlates with the customer satisfaction going down through the year.

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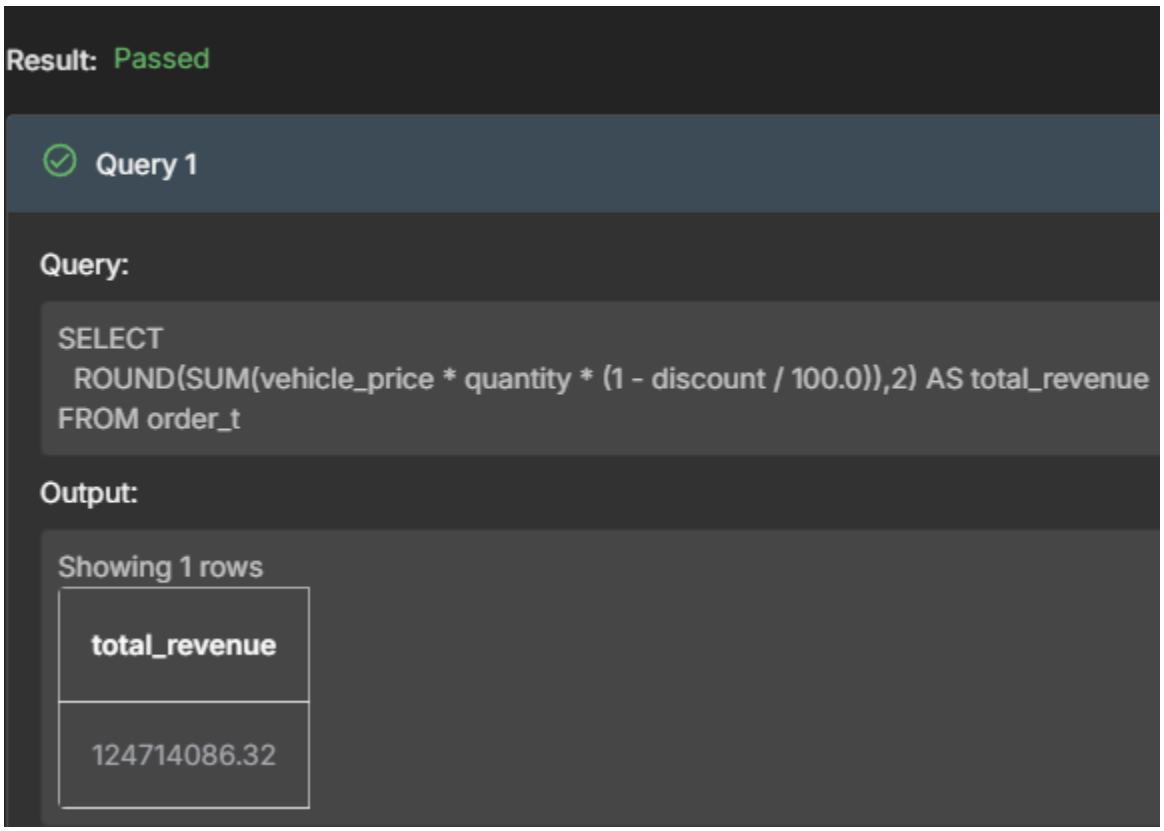
Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

Query 1

```
SELECT
   ROUND(SUM(vehicle_price * quantity * (1 - discount / 100.0)),2) AS total_revenue
FROM order_t
```

Output



Query 2

```
SELECT
   'Q' || quarter_number AS Quarter,
   ROUND(total_revenue, 2) AS Quarter_revenue,
   ROUND(
      (total_revenue - LAG(total_revenue) OVER (ORDER BY quarter_number))
```

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```
    / LAG(total_revenue) OVER (ORDER BY quarter_number) * 100
  , 2) AS Revenue_change_pct
FROM (
  SELECT
    quarter_number,
    SUM(vehicle_price * quantity * (1 - discount / 100.0)) AS total_revenue
  FROM order_t
  GROUP BY quarter_number
) ;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT
    'Q' || quarter_number AS Quarter,
    ROUND(total_revenue, 2) AS Quarter_revenue,
    ROUND(
        (total_revenue - LAG(total_revenue) OVER (ORDER BY quarter_number))
        / LAG(total_revenue) OVER (ORDER BY quarter_number) * 100
    , 2) AS Revenue_change_pct
FROM (
    SELECT
        quarter_number,
        SUM(vehicle_price * quantity * (1 - discount / 100.0)) AS total_revenue
    FROM order_t
    GROUP BY quarter_number
)
```

Output:

Showing 4 rows

Quarter	Quarter_revenue	Revenue_change_pct
Q1	39421580.16	
Q2	32715830.34	-17.01
Q3	29229896.19	-10.66
Q4	23346779.63	-20.13

Observations and Insights:

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- Total revenue has decreased over the four quarters by 41% total.
- The rate of decrease of revenue (percentage) from last quarter is highest from Q3 to Q4 - 20.13 % - this is a worst quarter over quarter decrease.
- The rate of decrease of revenue (percentage) from last quarter is lowest from Q2 to Q3- 10.66% - so this is a little bit better trend from Q2 to Q3.

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Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT
    'Q' || quarter_number AS Quarter,
    ROUND(SUM(vehicle_price * quantity * (1 - discount / 100.0)), 2) AS Total_Revenue,
    COUNT(order_id) AS Orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

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Result: Passed

Query 1

Query:

```
SELECT
   'Q' || quarter_number AS Quarter,
   ROUND(SUM(vehicle_price * quantity * (1 - discount / 100.0)), 2) AS Total_Revenue,
   COUNT(order_id) AS Orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

Quarter	Total_Revenue	Orders
Q1	39421580.16	310
Q2	32715830.34	262
Q3	29229896.19	229
Q4	23346779.63	199

Observations and Insights:

- Both total revenue and number of orders decreased in every quarter.
- The revenue seems proportional to the number of orders. The decrease in number of orders has a direct relationship with the revenue.
- Decreasing customer satisfaction may have caused the decrease in orders and the revenue.

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Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

Note: The '`discount`' field in the '`order_t`' table is stored as a percentage, i.e. `0.6` represents a discount of `0.6%`, not `60%`.

```
SELECT
   credit_card_type AS Credit_Card_Type,
   ROUND(AVG(discount), 2) AS Avg_Discount
FROM order_t
INNER JOIN customer_t USING (customer_id)
GROUP BY credit_card_type
ORDER BY Avg_Discount DESC;
```

Output:

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Result: Passed

Query 1

Query:

```
SELECT
   credit_card_type AS Credit_Card_Type,
   ROUND(AVG(discount), 2) AS Avg_Discount
FROM order_t
INNER JOIN customer_t USING (customer_id)
GROUP BY credit_card_type
ORDER BY Avg_Discount DESC
```

Output:

Showing first 10 rows out of 16 rows

Credit_Card_Type	Avg_Discount
laser	0.64
mastercard	0.63
visa-electron	0.62
maestro	0.62
instapayment	0.62
china-unionpay	0.62
americanexpress	0.62
switch	0.61

Observations and Insights:

- The average discount offered for laser credit cards is highest - 0.64%

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- The average discount offered by Diners-club-international is 0.58 % which is the lowest.
- The credit card discount is a very narrow range from 0.58% to 0.64%. There is not much difference in the discount offered.

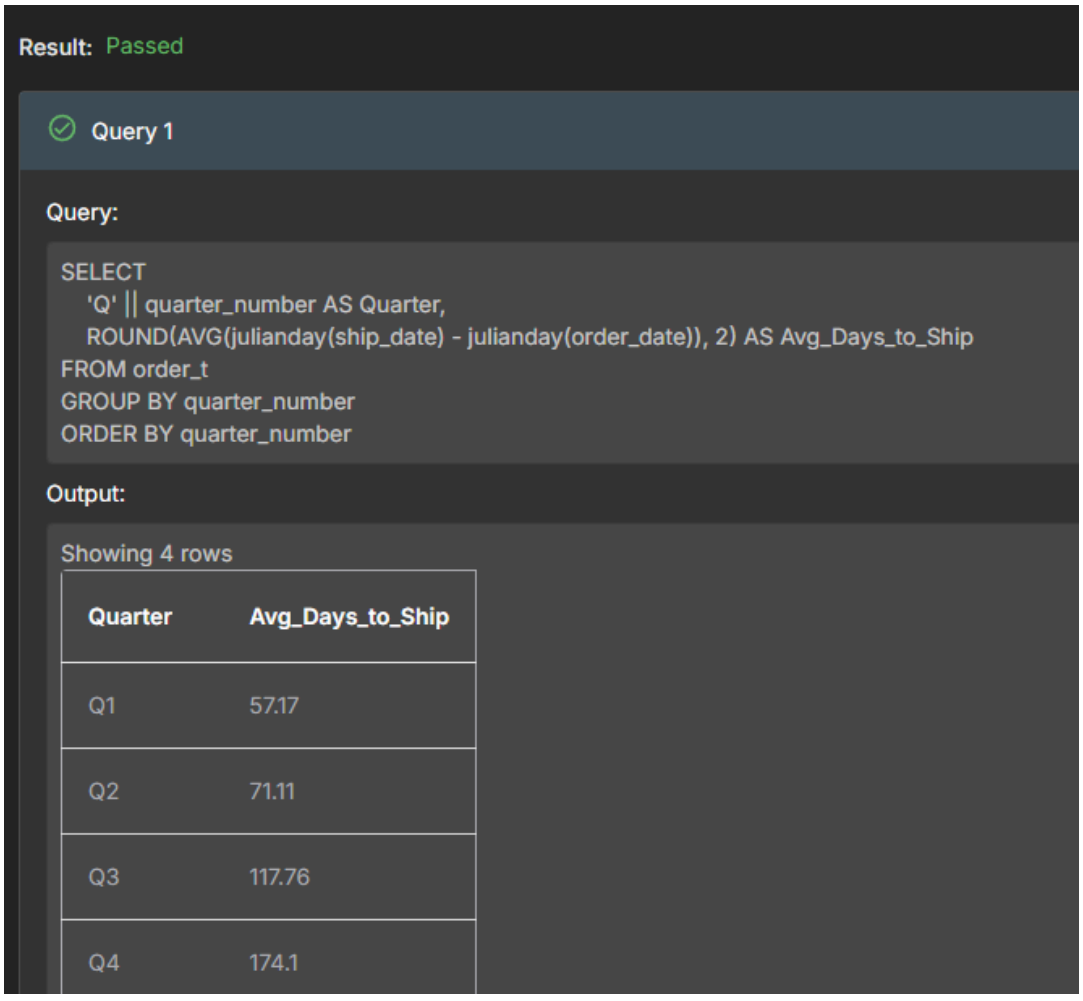
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Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT
   'Q' || quarter_number AS Quarter,
   ROUND(AVG(julianday(ship_date) - julianday(order_date)), 2) AS Avg_Days_to_Ship
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

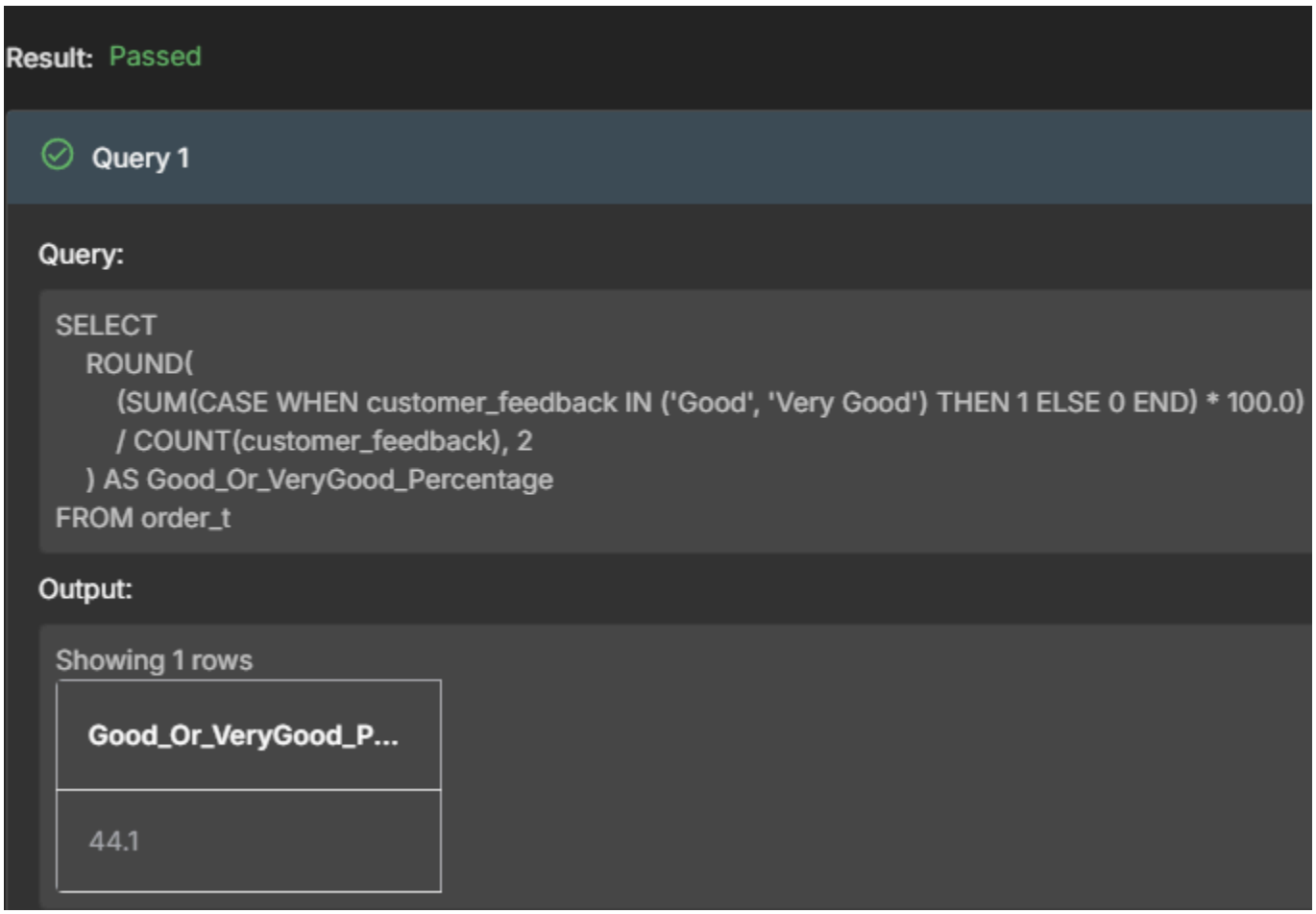
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Observations and Insights:

- The average time taken to ship the placed orders has increased over the four quarters.
- Q1 had 57.17 days for shipping, whereas Q4 had 174.1 days. Shipping time has increased almost 3 times!
- The increasing shipping time may be one cause of customer dissatisfaction!

Query for calculating % good feedback-



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Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
124.71 M	1000	994	3.14
Last Quarter Revenue	**Last quarter Orders**	**Average Days to Ship**	**% Good Feedback**
23.35 M	199	98	44.1%

Note: These values must be derived using SQL queries. Some of them may have already been obtained while answering previous questions.

Business Recommendations

- New Wheels had 124.71 M as the total revenue in the year. Customer satisfaction, number of orders and revenue are all showing a decreasing trend over the four quarters. Revenue has dropped by almost 41% from Q1 to Q4.
- The shipping time has shown an increasing trend over the year, and has tripled in a year. This may be a reason for the customer satisfaction and number of orders going down. These are directed related to the revenue. So, order processing and shipping delays need to be corrected by New Wheels.
- There are some US states that have only one customer. They are: Maine, Vermont, Wyoming. Maybe New Wheels can shut down sales operations in these states.
- New Wheels should focus on the sales in the top states which are : California, Texas, Florida, New York and District of Columbia.
- New Wheels should focus on the sales with top vehicle makers which are : Chevrolet, Ford, Toyota, Pontiac, Dodge.

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